



**UNIVERSIDAD DE SAN CARLOS DE GUATEMALA
CENTRO UNIVERSITARIO DE OCCIDENTE
DEPARTAMENTO DE ESTUDIOS DE POSTGRADOS
"MAESTRIA EN INGENIERIA ESTRUCTURAL Y
SISMORRESISTENTE "
CURSO: INGENIERÍA SÍSMICA
MSc. ING. JAVIERA YOLANDA MALDONADO DE LEÓN**

"EVALUACIÓN FINAL PRACTICA "

ALUMNO:
MARLON IVAN CARRETO RIVERA **201230088**

FECHA

18 DE AGOSTO 2026

EXAMEN FINAL PRÁCTICO — INGENIERÍA SÍSMICA
PROBLEMA 1 — ESPECTRO NORMATIVO AGIES NSE 2-
2024

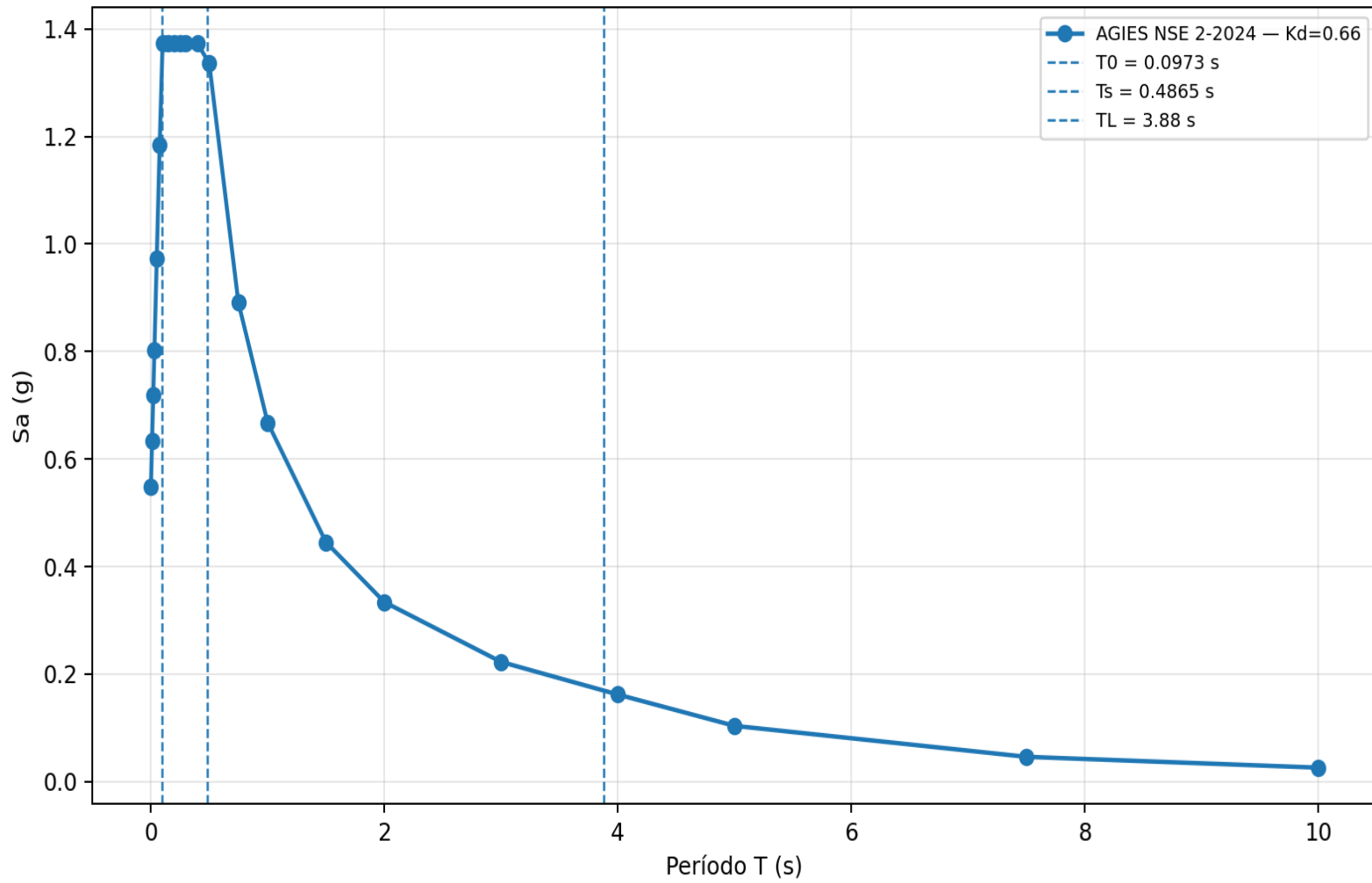
Estudiante	Marlon Ivan Carreto Rivera
Registro	201230088
Fila asignada	9
Tipo de suelo	D
Problema 1	Siquinalá, Escuintla, Guatemala

Parámetro	Valor
Municipio	Siquinalá
Departamento	Escuintla
Índice de sismicidad I_0	4.2
Clase de suelo	D
S_{cr}	2.08
S_{1r}	1.012
T_L	3.88
Velocidad básica de viento	100
K_d normativo adoptado	0.66
S_{cd}	1.3728
S_{1d}	0.66792
T_s	0.486538462
T_0	0.097307692
AMS_d	0.54912
S_{vd}	0.27456

Parámetro	Valor	Unidad
Kd	0.66	
Scd	1.3728	g
S1d	0.66792	g
Ts	0.486538462	s
T0	0.097307692	s
TL	3.88	s

T (s)	Sa (g)
0.000	0.5491
0.010	0.6338
0.020	0.7184
0.030	0.8031
0.050	0.9724
0.075	1.1840
0.100	1.3728
0.150	1.3728
0.200	1.3728
0.250	1.3728
0.300	1.3728
0.400	1.3728
0.500	1.3358
0.750	0.8906
1.000	0.6679
1.500	0.4453
2.000	0.3340
3.000	0.2226
4.000	0.1620
5.000	0.1037
7.500	0.0461
10.000	0.0259

Espectro normativo — Siquinalá, Escuintla — Suelo D



PROBLEMA 2, FLORIDA, USA



Address:
Florida

ASCE Hazards Report

Standard: ASCE/SEI 7-22

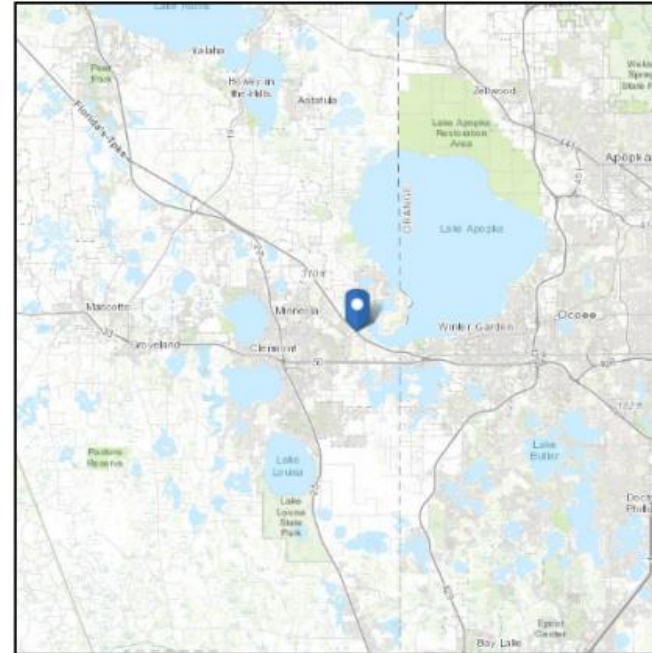
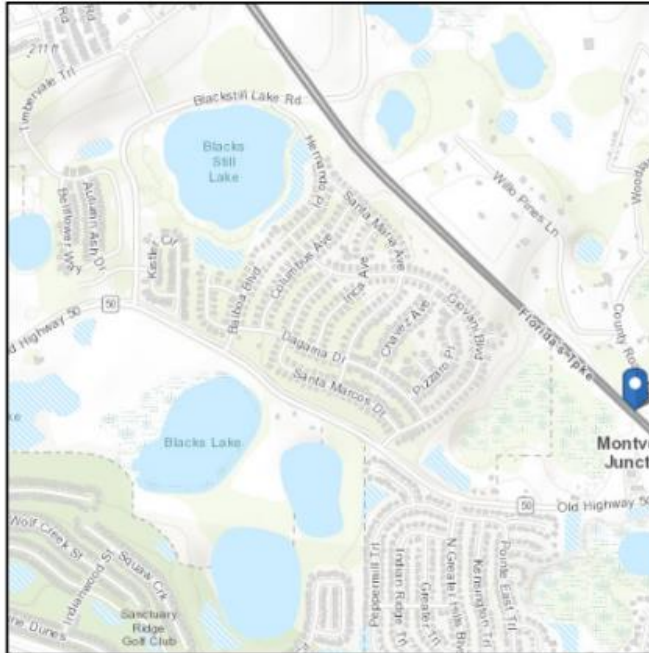
Risk Category: III

Soil Class: D - Stiff Soil

Latitude: 28.566045

Longitude: -81.688649

Elevation: 86.29932454307435 ft
(NAVD 88)

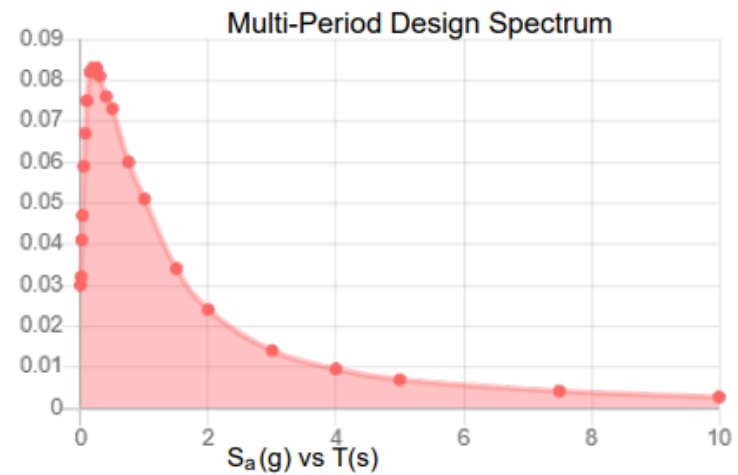
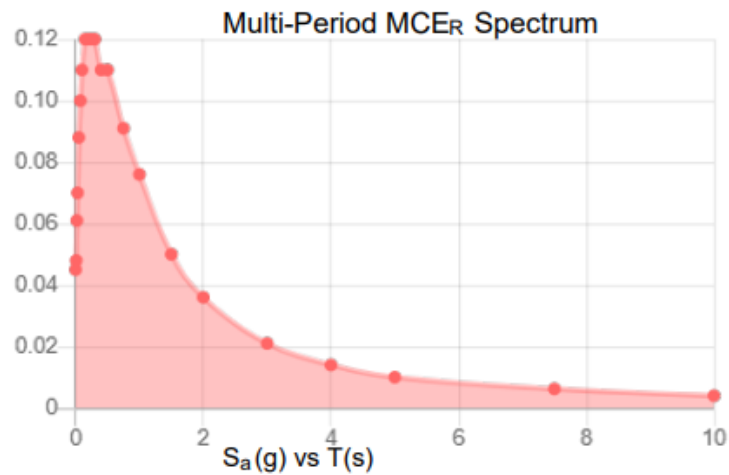


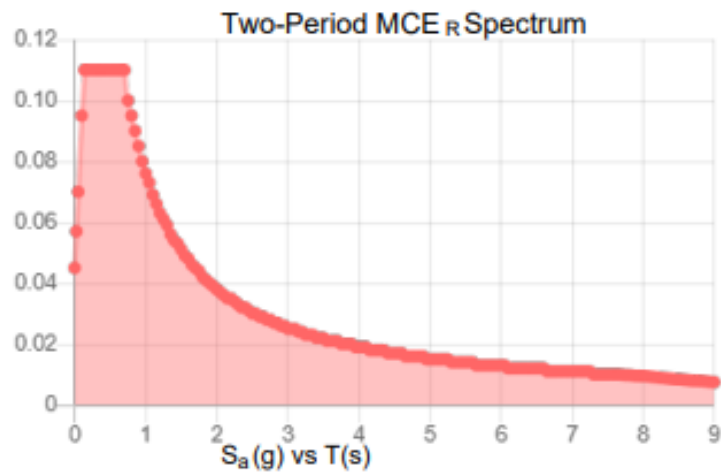
Site Soil Class: D - Stiff Soil

Results:

PGA_M :	0.041	T_L :	8
S_{MS} :	0.11	S_S :	0.084
S_{M1} :	0.076	S_1 :	0.036
S_{DS} :	0.075	V_{S30} :	260
S_{D1} :	0.051		

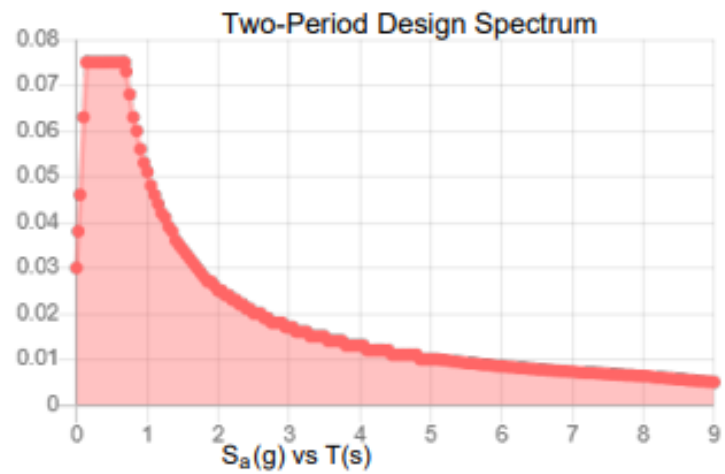
Seismic Design Category: A





MCE_R Vertical Response Spectrum

Vertical ground motion data has not yet been made available by USGS.



Design Vertical Response Spectrum

Vertical ground motion data has not yet been made available by USGS.